2.5 Solving Compound Inequalities

Name _____ Period ____ Date _____

Algebra I 2.5 · Student sheet

I can…

- write and graph compound inequalities.
- solve compound inequalities that use “and.”
- solve compound inequalities that use “or” and use them in real life.

Vocabulary · fill in the missing word

Compound inequality — two inequalities _____ by “and” or “or”

“And” — the solutions satisfy BOTH _____ (the overlap, often written $a < x < b$)

“Or” — the solutions satisfy AT LEAST ONE _____ (two pieces)

Part 1 · Temperature ranges

Fish tank: the water must stay between 72°F and 82°F, inclusive.

Inequality: _____ Graph:

Pool: it closes if the water is below 60°F or above 95°F.

Inequality: _____ Graph:

Reading	70°F	75°F	82°F	90°F
Fish tank OK?				
Pool closed?				

Part 2 · Notes: fill in the blanks

“And”: _____ inequalities must be true. Graph the _____.

“Or”: at least _____ inequality must be true.

Solve a three-part inequality by doing the same thing to _____ parts.

Part 3 · Examples (work them with the class)

Example 1

Write each as a compound inequality.

a. all real numbers greater than −3 and less than 2

b. all real numbers less than −1 or at least 4

Example 2

Solve $-4 < x + 3 \leq 9$ and graph.

Example 3

Solve $-8 \leq 2x - 4 < 6$.

Example 4

Solve $3x + 1 < -5$ or $2x - 3 \geq 7$ and graph.

Part 4 · Partner practice

Partner 1

Solve $5 \leq x - 2 < 10$.

Partner 2

Solve $-x > 4$ or $x + 1 > 6$.